

Introduction to AI Safety, Ethics, and Society

Artificial Intelligence Fundamentals

Artificial Intelligence

Definitions, history, and types

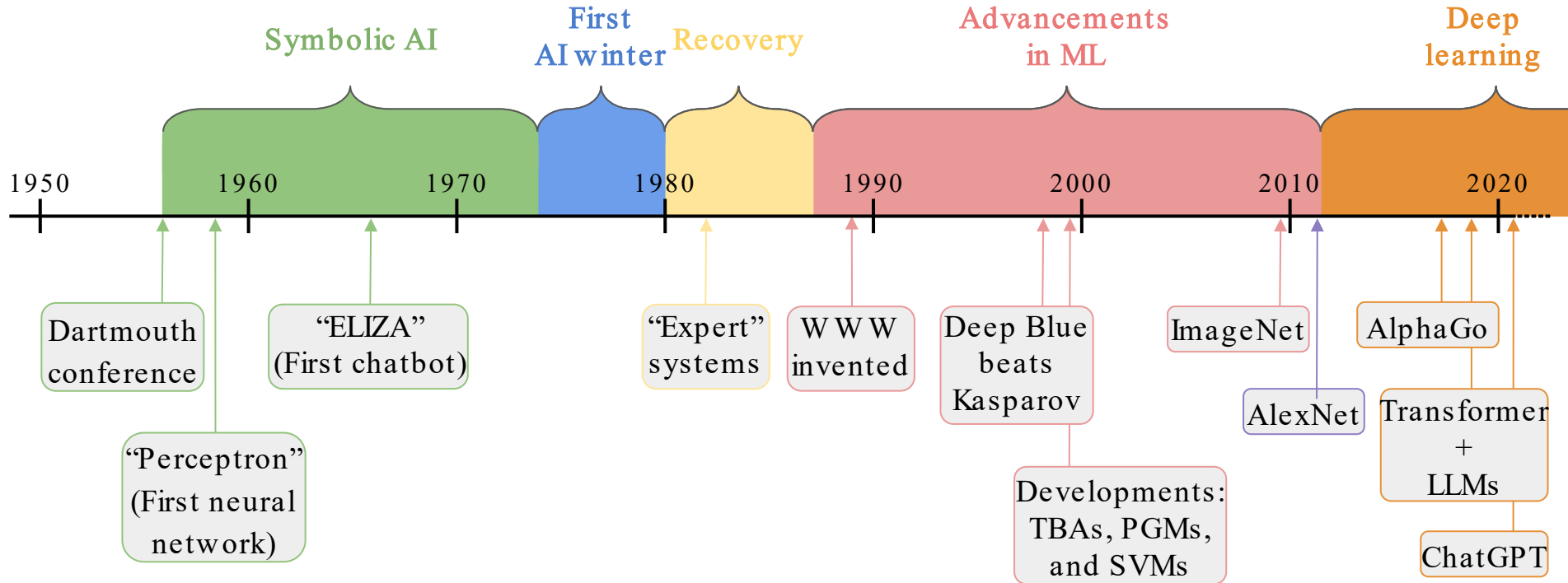
Definition of AI

There is no single universally-accepted definition of “artificial intelligence.”
It is used to describe different (but related) ideas:

- Branch of computer science
- Type of machine
- Tool
- Component of a business model
- Philosophical idea

Our definition: computer systems that perform tasks typically associated with intelligence such as problem solving, decision making, forecasting.

Modern History of AI



Types of AI

Narrow AI: capable of performing specific tasks only.

Artificial General Intelligence (AGI): capable of performing many cognitive tasks across multiple domains, at human or superhuman levels.

Human-level AI (HLAI): capable of doing virtually everything humans do.

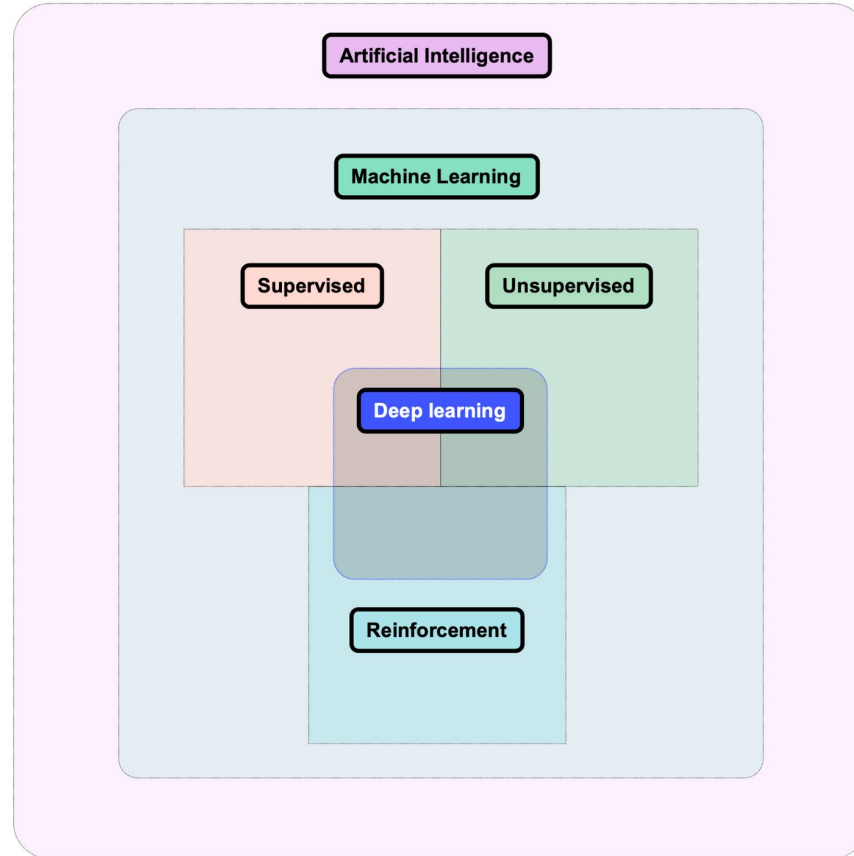
Transformative AI (TAI): systems with a dramatic impact on the world at the level of the industrial revolution.

Superintelligence (ASI): capable of outclassing human performance on virtually all intellectual tasks.

Machine Learning

The models, tasks, data, and types of learning

Machine Learning Definitions



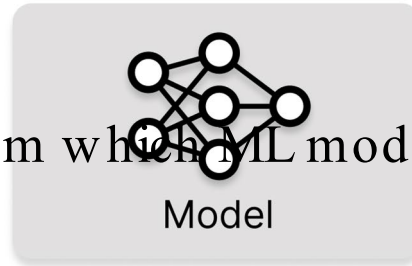
Machine Learning Definitions

Machine learning (ML): developing computer systems that can learn directly from data without following explicit pre-set instructions.

Algorithm: a procedure (recipe) for accomplishing a task.

ML model: an algorithm designed to learn automatically from data by identifying patterns and relationships to make predictions or decisions.

Training data: the data from which ML models learn.



Model Access

Model developers can grant different levels of **access** to the public.

Closed Models: Operate in a closed system, not accessible for external use or modification, often used internally by organizations.

API Models: Accessible through an interface with usage restrictions, allowing users to query the model without viewing or modifying the underlying model.

Open Models: Fully accessible and modifiable, often part of open-source projects, allowing users to use, modify, and distribute freely.

Computational Resources

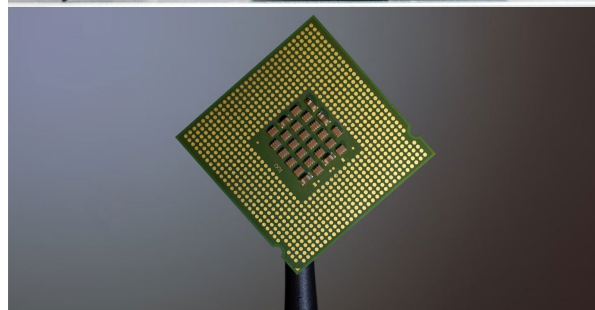
Compute in ML: Essential for handling the large datasets and complex calculations in machine learning, with GPUs and CPUs being primary examples.

GPUs (Graphics Processing Units):

- Parallel processing

CPUs (Central Processing Units):

- Sequential processing



Tasks, Inputs, Outputs

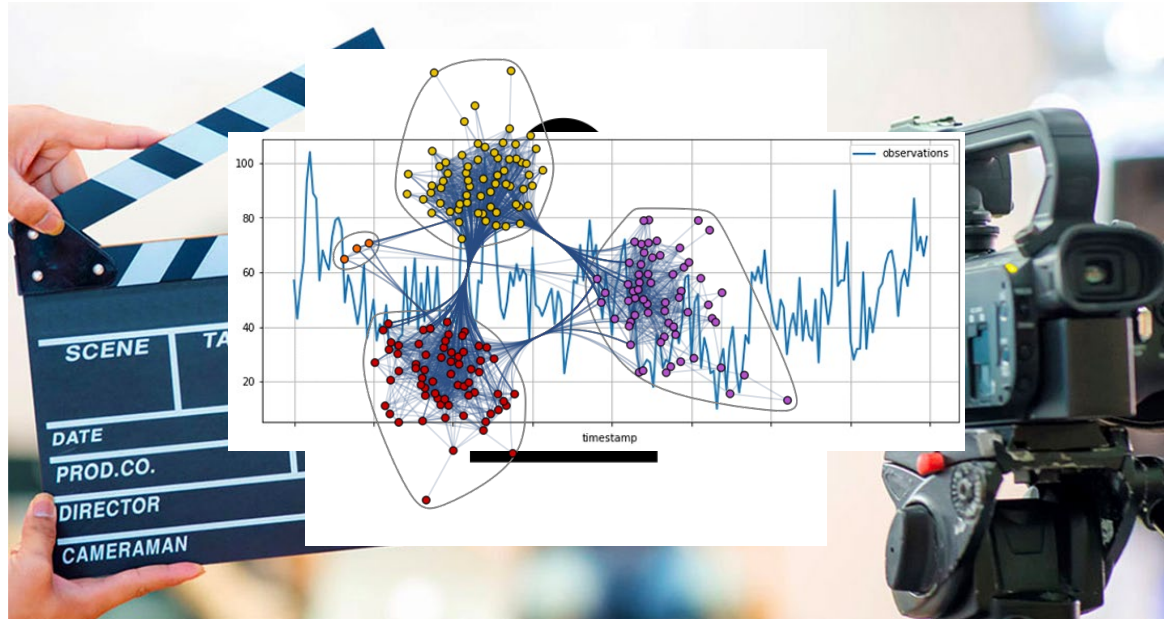
1. **Task:** What is the primary goal of the ML model?
1. **Inputs:** What data does the ML system receive?
1. **Outputs:** What does the ML system produce?

Types of Input Data

Modality: a type of input to an ML model

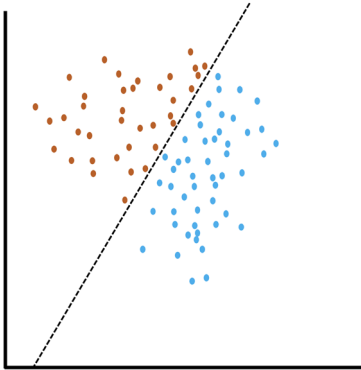
Major modalities in machine learning:

- Tables
- Text
- Images
- Video
- Audio
- Time series
- Graphs
- Set values

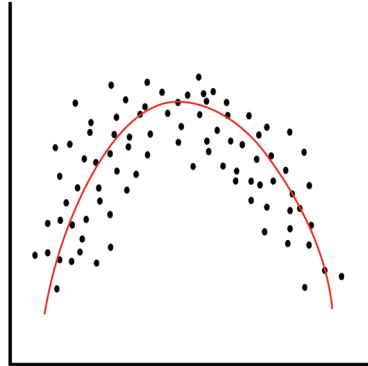


Key ML Tasks

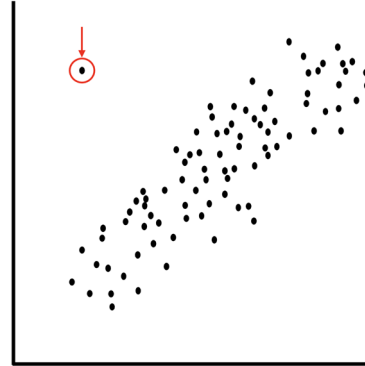
Classification



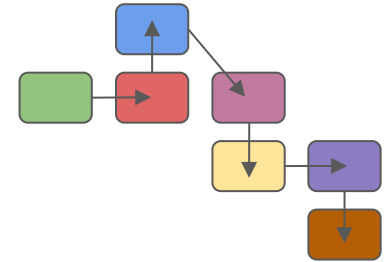
Regression



**Anomaly
detection**



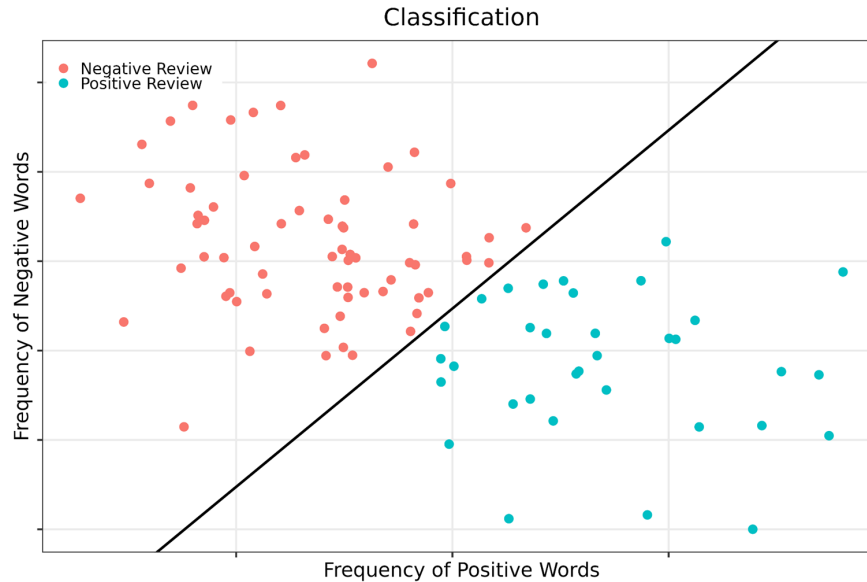
**Sequence
modelling**



Task 1:

Classification

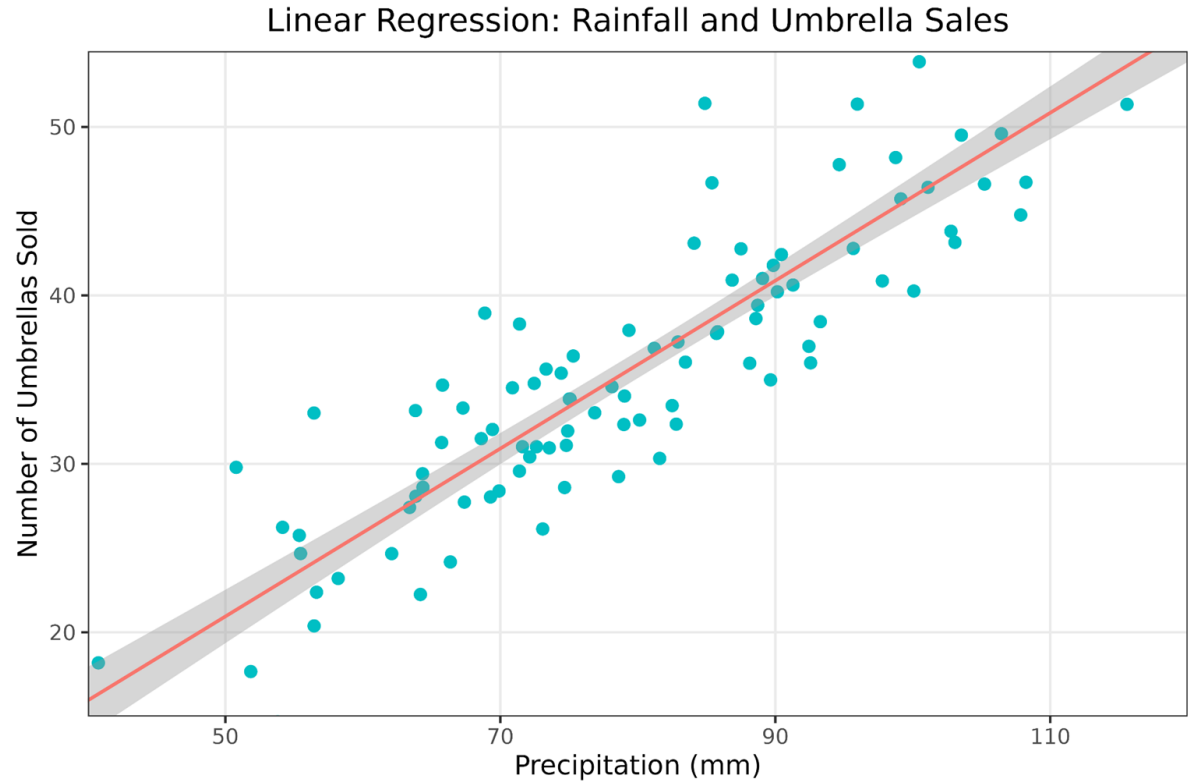
Classification: predicting categories or “classes” using features of an input data point.



Task 2:

Regression

Regression:
predicting numerical
outputs using
features of input
data.



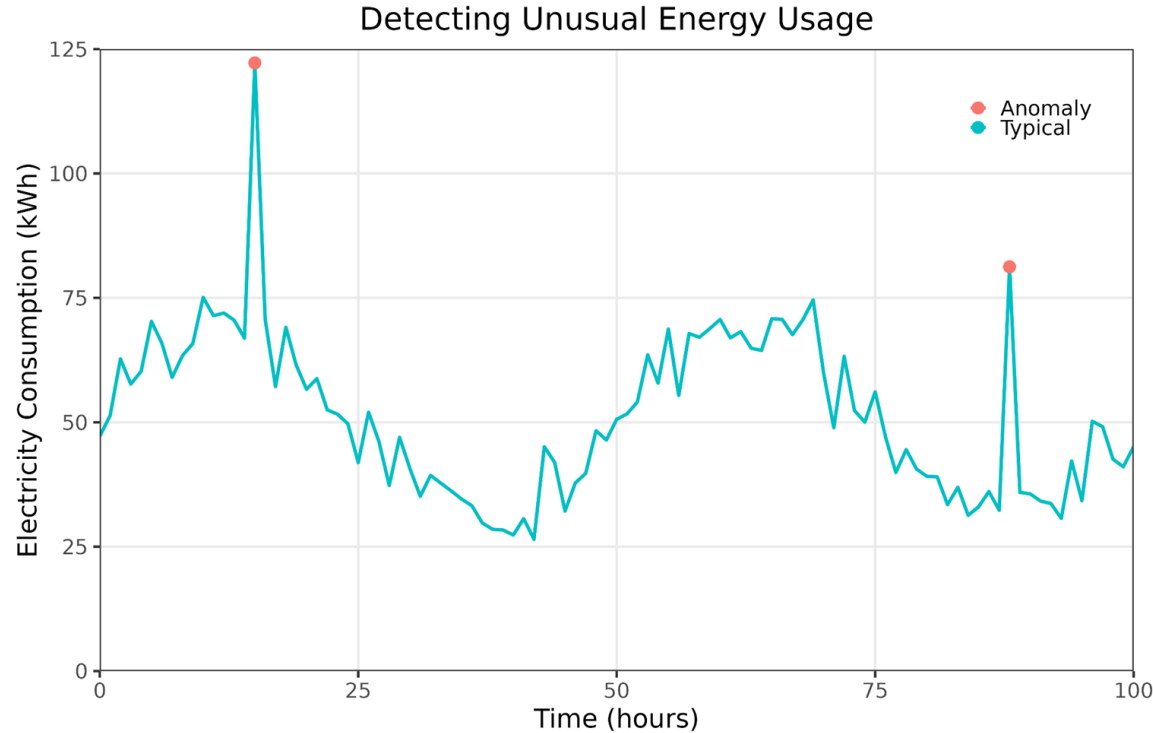
Task 3:

Anomaly Detection

Anomaly detection:
identifying outliers or
abnormal data points.

Example: black swan
detection.

Black swans: events that
are rare and unpredictable,
yet impactful



Sequence Modeling

Sequence modeling: analyzing and predicting patterns in sequential data.

Generative modeling: creating new data that resembles the input data;
ChatGPT is a “generative model”

Components of the ML Pipeline

ML pipeline: a series of interconnected steps in the process of developing a ML model.

Steps:

- Data collection
- Labeling
- Choosing architecture
- Training
- **Evaluation**
- Deployment

Three Types of Machine Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

Type 1:

Supervised Learning

Supervised learning: learning from labeled data.

- Learning relationships between input data and output labels
- Labels are usually generated by humans

Example: Species identification

Advantage: high accuracy and reliability for large and well-labeled dataset.

Disadvantage: worse performance for loosely-defined tasks, like poetry or image generation; may also require manual labeling.



Unsupervised Learning

Unsupervised learning: learning from unlabeled data.

- Discovering and identifying patterns within a dataset to learn the relationships between variables
- Also known as “self-supervised learning,” because the model may use these patterns to generate supervisory signals for itself

Examples:

- Language models
- Image inpainting



Type 2:

Unsupervised Learning



Type 3:

Reinforcement Learning

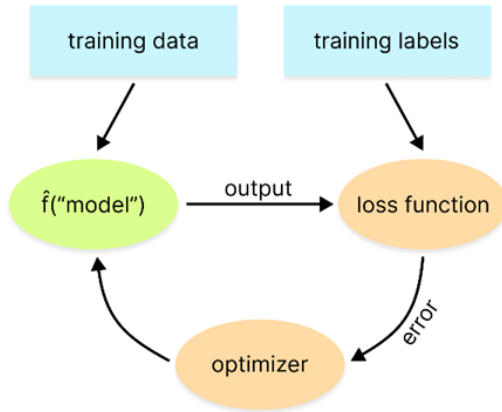
Reinforcement learning (RL): learning from agent-gathered data.

- AI agents making decisions and improving their performance based on responses to their environment

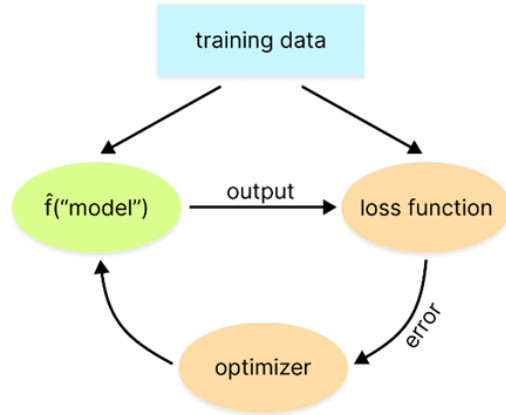
Example: robots navigating environments through trial-and-error.

Review of Machine Learning Types

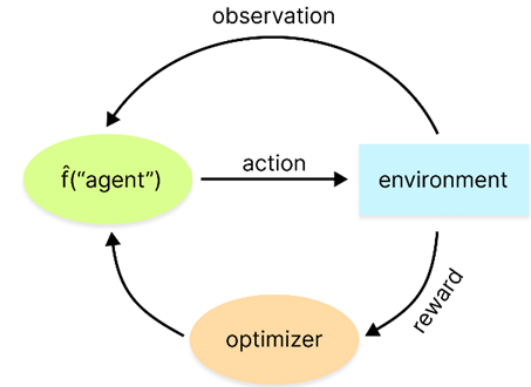
Supervised Learning



Unsupervised Learning



Reinforcement Learning



Recap

AI systems are computer systems that perform tasks typically associated with intelligence such as problem solving, decision making, forecasting. They can be classified according to their capabilities.

Machine Learning systems can learn directly from data without following explicit pre-set instructions. They are used for tasks like classification, regression, anomaly detection, and sequence modeling.

ML requires data, architecture, training it, evaluating, deploying.

ML can use supervised, unsupervised, and reinforcement learning.

Roadmap

1. Artificial Intelligence: *Definitions, history, and types.*

1. Machine Learning: *Models, tasks, data, and learning types.*

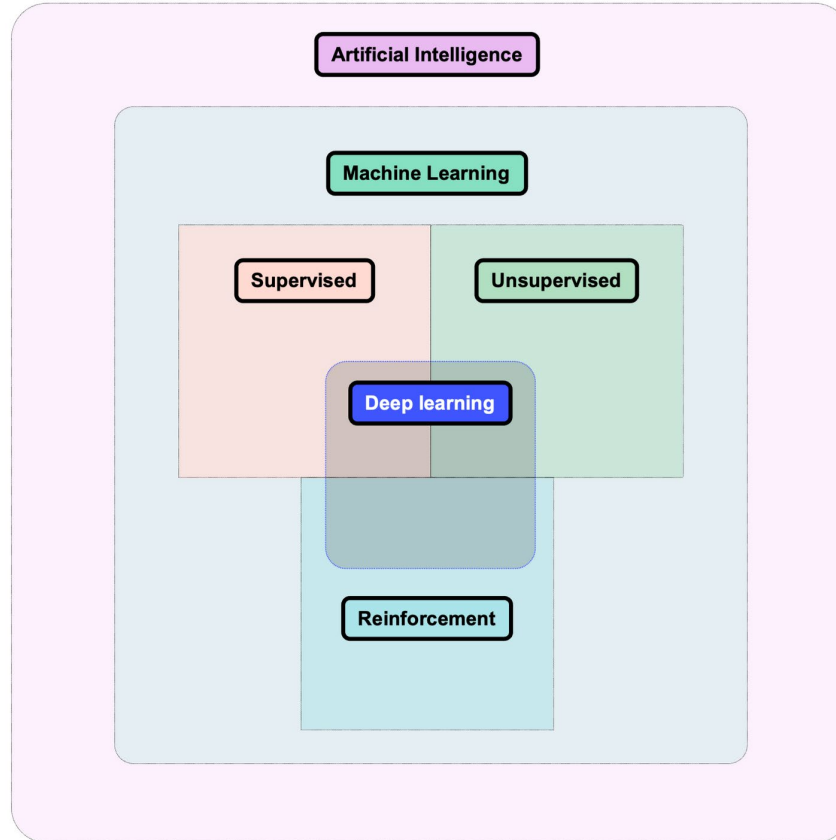
2. Deep Learning: *The preeminent paradigm.*

1. Scaling Laws: *Many AI systems improve predictably with scale.*

Deep Learning

The preeminent paradigm

Introduction to Deep Learning



Introduction to Deep Learning

Deep learning (DL):

- Uses **neural networks**: layers of interconnected nodes that transform inputs into outputs, inspired by biological neurons.
- Extracts useful patterns from large quantities of data
- Useful for learning hierarchical representations of the world

Why Deep Learning Matters

Performance:

- Superhuman in some domains (though not reliably so)
- Examples: image recognition, chess, and Go

Real-world usefulness:

- Wide variety of useful applications
- Examples: healthcare, social media, and autonomous vehicles

Scalability:

- DL models are highly scalable
- Performance tends to improve with more data, rather than tapering off

Chapter Recap

Artificial Intelligence: Definitions, historical development, and types.

Machine Learning: Models, tasks, data, and learning types.

Deep Learning: Fundamental concepts and applications.